

Lego Set Explorer Project ReadME

Project Overview :

This project involves building an interactive LEGO Set Explorer in Microsoft Power BI. The report allows users to browse and filter over 4,000 LEGO sets by theme group, theme and age range, view key metrics at a glance, and explore detailed information (including the product image, price, year and piece count) for any individual set selected from the data table.

Objective :

- Load and clean a LEGO dataset using Power Query Editor
- Engineer categorical features using Conditional Columns
- Create DAX measures for aggregated metrics
- Design an interactive, image-rich report with slicers, a filterable table and a dynamic detail panel
- Control visual interactions to prevent unwanted cross-filtering

Dataset Used :

Set_ID	Official LEGO item number
Name	Name of the LEGO set
Year	Release Year
Theme	LEGO theme the set belongs to
Sub-theme	Sub-theme within the theme
Theme Group	Overall group the theme belongs to
Category	Type of set
Pieces	Number of pieces in the set
Age	Minimum age recommended
Price	US retail price at launch
Image URL	Full size image of the set

Age Range	Conditional Column (1 to 4, 5 to 9, 10 to 17, Over 18)
Price Range	Conditional Column (\$ to \$\$\$\$\$) for categorical filtering

Steps Performed :

1. Data Loading & Transformation

2. Data Modelling :

Added Conditional Columns - Age Range and Price Range

3. DAX Measures (Data Analysis Expressions) -

- Total Sets = DISTINCTCOUNT(lego_sets[set_id])
- Total Groups = DISTINCTCOUNT(lego_sets[themeGroups])
- Average Price = AVERAGE(lego_sets[price])
- Average Age = AVERAGE(lego_sets[age])
- Average Pieces = AVERAGE(lego_sets[pieces])

4. Dashboard Design.

Added Slicers.

Tooltips to preview set image by hovering on the table.

'Reset Filters' button by assigning Bookmark action.

Key Insights :

- The dataset covers 4,385 unique LEGO sets across 15 theme groups.
- The average set contains 411 pieces and is priced at \$45.

- The Price Range buckets (\$-\$\$\$\$\$) enable quick identification of premium vs. entry-level sets.
- Age Range slicing reveals which themes target specific age groups, supporting both retail strategy and consumer research.

Dashboard Preview :

- Header row: LEGO logo + three KPI cards (Total Sets, Average Pieces, Average Price)
- Filter row: Three dropdown slicers (Theme Group, Theme, Age Range)
- Main panel: Scrollable data table (left) + Set Detail panel with image and info cards (right)
- Action button to reset filters.